

DIS App Development Guide

Prerequisites

Before starting development, ensure you have the following:

- **Git & Node.js / Next.js Environment:** Ensure **Git** and **Node.js** are installed on your local machine.
- **Developer Portal Account:** An active account on the DIS Developer Portal for application registration and credential generation.

Steps

Use the given API Catalog to brainstorm a user flow and build a small application using **2–3 APIs from different departments**.

1. Clone the starter template

```
Shell
git clone
https://github.com/GovTech-SL/dis-demo-nextjs-starter-template.git

cd dis-demo-nextjs-starter-template/
```

2. Create a new branch

Branch naming convention: `app/<app-name>`

```
Shell
git checkout -b app/<your-app-name>
```

Example:

```
Shell
git checkout -b app/welfare-assistance
```

3. Choose the APIs

- Explore the available API catalogs, such as the **Developer Portal API Catalog**, the **Onboarded APIs Reference**, or the shared PDF guide.
- Select **2–3 APIs** across different departments/agencies (e.g., Ministry of Health, Welfare Benefits Board, or TVEC) to support your targeted integration flow.

4. Create an application in the Developer Portal

- Create a new application.
 - Use white list IP as **192.168.2.1**

Edit application Update your application details and configuration Exit to Applications

✓ Application details ✓ API selection 3 Configuration 4 Review

Advanced Configuration
Configure security and performance settings

Webhook URL (Optional)

Receive real-time notifications for events 0/255

IP Whitelist *

Restrict API access to specific IP addresses (comma separated)

Data Residency

☒ **Sri Lanka Only**
All data stored within Sri Lanka borders

☐ **Regional (South Asia)**
Data stored in South Asian region

☐ **Global**
Data stored globally for optimal performance

< Previous Next: Review >

- Add the selected APIs.
- Capture the generated:
 - `DIS_CLIENT_ID`
 - `DIS_API_KEY`

5. Configure the environment

Add the credentials to the `.env` file in the starter template:

```
Shell
cp .env.example .env
```

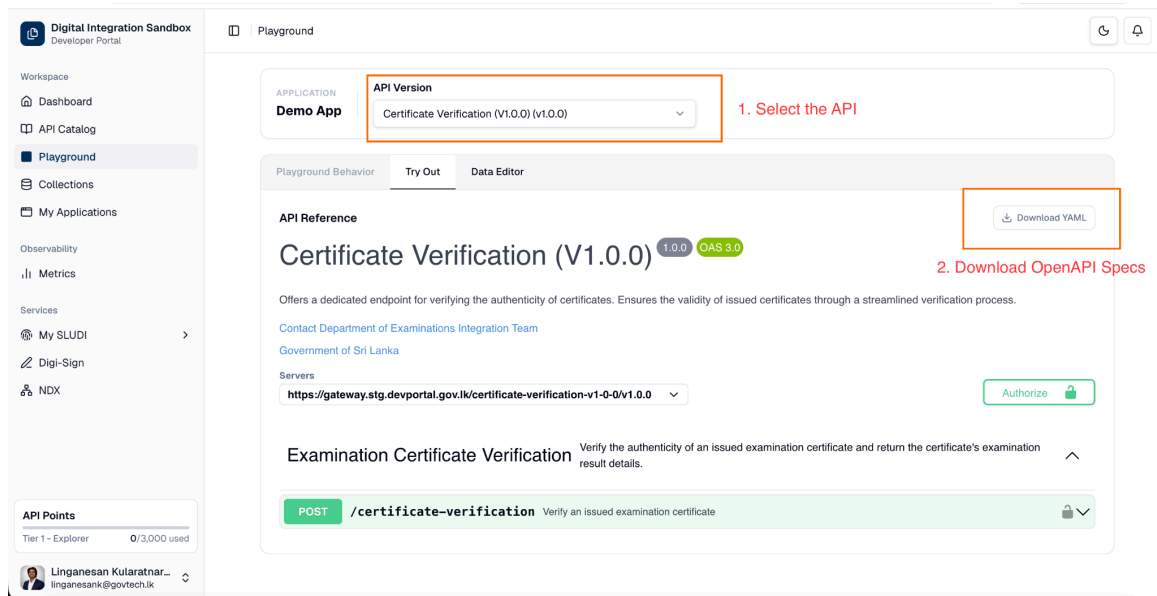
None

DIS_CLIENT_ID=<your-client-id>

DIS_API_KEY=<your-api-key>

6. Download OpenAPI Specifications and Organize Data

Download the OpenAPI YAML file for each selected API and save them to the `/openAPI` directory in your starter template.



Including these specifications provides your AI coding tools with essential context, enabling them to better understand API endpoints, request/response schemas, and integration logic.

Additionally, organize your test data by creating a `seed-data/` folder within the `/openAPI` directory. Structure your CSV files as shown below to ensure clarity and consistency

Click to go forward, hold to see history

Developer Portal

Workspace

Dashboard

API Catalog

Playground

Collections

My Applications

Observability

Metrics

Services

My SLUDI

Digi-Sign

NDX

API Points

Tier 1 - Explorer 0/3,000 used

Unganesan Kularatnar...

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Playground

APPLICATION

Demo App

API Version

Certificate Verification (V1.0.0) (v1.0.0)

Playground Behavior

Try Out

Data Editor

Download Seed Data

TABLES

certificates 7

certificate_subject... 47

Import CSV

Download CSV

Reset to Seed

+ Add Row

One by One

id	stream	zScore	IndexNo	examName
cert-001	BIOSYSTEMS TECHNOLOGY	0.1462	7748381	GENERAL CERTIFICATE OF EDUCATION ADVANL
cert-002	PHYSICAL SCIENCE	1.2845	7748399	GENERAL CERTIFICATE OF EDUCATION ADVANL
cert-003	null	null	8363462	GENERAL CERTIFICATE OF EDUCATION ORDINA
cert-004	BIOLOGICAL SCIENCE	+0.4064	7283842	GENERAL CERTIFICATE OF EDUCATION ADVANL
cert-005	COMMERCE	+1.7821	6643011	GENERAL CERTIFICATE OF EDUCATION ADVANL
cert-006	null	null	5512098	GENERAL CERTIFICATE OF EDUCATION ORDINA
cert-007	null	null	4401786	GENERAL CERTIFICATE OF EDUCATION ORDINA

None

/openAPI

```

├─ welfare-enumeration-api.yaml
├─ pensioner-eligibility-services.yaml
├─ seed-data/
│   └─ welfare-enumeration-api/
│       ├── data_1.csv
│       └─ data_2.csv
└─ pensioner-eligibility-services/
    ├── data_1.csv
    └─ data_2.csv

```



AI Code Editor Prompt

Create folders under seed-data/ for each API to store the csv files for seed data.

6. Develop Application

AI Code Editor Prompt

Create the Application with minimal UI and showing API calls in proper way.

7. Update the Seed Data to test end-to-end flow

AI Code Editor Prompt

Update the seed data to test the user-flow end to end.

Playground BehaviorTry OutData Editor

Import updated seed data CSV file for that table

TABLES

certificates7

certificate_subject_...47

certificates7 rows

Import CSV

Download CSV

Reset to Seed

+ Add Row

id	stream	zScore	indexNo	examName
cert-001	BIOSYSTEMS TECHNOLOGY	0.1462	7748381	GENERAL CERTIFICATE OF EDUCATION ADVAN...
cert-002	PHYSICAL SCIENCE	1.2845	7748399	GENERAL CERTIFICATE OF EDUCATION ADVAN...
cert-003	null	null	8363462	GENERAL CERTIFICATE OF EDUCATION ORDINA...
cert-004	BIOLOGICAL SCIENCE	+0.4064	7283842	GENERAL CERTIFICATE OF EDUCATION ADVAN...
cert-005	COMMERCE	+1.7821	6643011	GENERAL CERTIFICATE OF EDUCATION ADVAN...
cert-006	null	null	5512098	GENERAL CERTIFICATE OF EDUCATION ORDINA...
cert-007	null	null	4401786	GENERAL CERTIFICATE OF EDUCATION ORDINA...

Select the Table to update

Playground Behavior	Try Out	Data Editor			
TABLES	certificates 7 rows		Imported 7 rows into certificates.		Import CSV Download CSV Update Changes Reset to Seed
	+ Add Row		Update the changes		
certificates 7	id	stream	zScore	indexNo	examName
certificate_subject_... 47	cert-001	BIOSYSTEMS TECHNOLOGY	0.1462	7748381	GENERAL CERTIFICATE OF EDUCATION ADVAN...
	cert-002	PHYSICAL SCIENCE	1.2845	7748399	GENERAL CERTIFICATE OF EDUCATION ADVAN...
	cert-003	null	null	8363462	GENERAL CERTIFICATE OF EDUCATION ORDINA...
	cert-004	BIOLOGICAL SCIENCE	+0.4064	7283842	GENERAL CERTIFICATE OF EDUCATION ADVAN...
	cert-005	COMMERCE	+1.7821	6643011	GENERAL CERTIFICATE OF EDUCATION ADVAN...
	cert-006	null	null	5512098	GENERAL CERTIFICATE OF EDUCATION ORDINA...
	cert-007	null	null	4401786	GENERAL CERTIFICATE OF EDUCATION ORDINA...

Best Practices for App Development

- **Isolate Credentials & Environments:** Never commit API keys or client secrets to source control. Always use environment variables (`.env`) and sandbox credentials during development.
- **Utilize Synthetic Data:** Perform all initial behavioral testing and flow validation in the sandbox environment using synthetic test data.
- **Leverage OpenAPI Specifications:** Keep OpenAPI specs updated in the `/openAPI` directory to assist AI coding assistants and ensure precise schema compliance.
- **Strict Application-Level Scoping:** Bind only the required APIs to your specific application profile in the Developer Portal to adhere to strict security boundaries.